

DISPOSITIONAL “WILL” IS “WANT” IN COLLOQUIAL JAKARTA INDONESIAN*

Anastasia Tsilia

Abstract. The verb *mau* ‘want’ can function as a future marker in Indonesian (Copley 2002, 2010; Jeoung 2020). In this use, it does not require an attitude holder and is compatible with inanimate subjects. Based on original fieldwork in Colloquial Jakarta Indonesian, I identify novel restrictions on *mau* as a future marker, arguing that it behaves as a weak positive polarity item: it cannot be directly negated, as negating *mau* negates ‘want’ rather than ‘will.’ Moreover, *mau* as ‘will’ is more restricted than the colloquial future marker *bakal*, requiring that there be “signs” of the future event. I propose that future *mau* requires its subject to be disposed at the time of evaluation to causing the future event, which I formalize as a presupposition on future *mau* using the notion of dispositional causation from Copley (2018). This study contributes to understanding *mau* as well as the synchronic connection between desire verbs and future modals, a diachronic pattern attested across languages (Bybee & Pagliuca 1987; Heine 2017 a.o.).

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1. Introduction

The verb *mau* ‘want’ in Indonesian also has a life as a future modal meaning ‘will’ (Copley 2002, 2010).¹ This paper presents original fieldwork conducted with a native speaker of Colloquial Jakarta Indonesian, and investigates several empirical puzzles related to the meaning of *mau* as ‘will’ (henceforth future *mau*). More specifically, I identify a novel negation puzzle, observing that future *mau* empirically patterns with weak Positive Polarity Items (henceforth PPI). I also characterize the empirical distribution of future *mau*, observing that it is restricted in interesting ways; future *mau* is a proximate future, and imposes a semantic requirement on its subject. I argue that it has a presupposition that the subject of future *mau* is disposed to causing the future event.

Verbs often appear bare in Colloquial Jakarta Indonesian, a sentence with no future marking being compatible with both a present and a past tense interpretation. It is the context that disambiguates between the two. Therefore, the following sentence with *nyanyi* ‘sing’ is compatible both with a present and with a past tense interpretation, but crucially not with a future tense one.

(1) Lia nyanyi.

Lia sing

‘Lia is singing/sang.’

The sentence above is compatible with contexts in (2a) and (2b), but not with the future context in (2c).

¹ See Jeoung (2020) for arguments in favor of categorial ambiguity of *mau*, although we need not fully commit to his analysis here. The empirical point made in this paper is that *mau* can be interpreted both as a desire verb and as a proximate future.

- (2) a. Context: Lia is singing now.
 b. Context: Lia sang a song and is not singing anymore.
 c. Context: Lia is going to sign.

Instead, the future is obligatorily marked. In Colloquial Jakarta Indonesian this is usually done with the marker *bakal* preceding the verb stem.²

(3) Context: What will Lia do at the event?

Lia *(bakal) nyani.

Lia will sing

‘Lia will sing.’

To receive a future tense interpretation in the sentence above, *bakal* is necessary, even if the context is such that we are clearly referring to the future. Another way to have a future tense interpretation without *bakal* is if there is a temporal phrase in the sentence referring to the future, as in the following:

(4) Tiga hari lagi, aku (bakal) main buluh barang sepupu aku.

Three day more I will play ball stuff cousing mine

‘In 3 days, I will play football with my cousin.’

In the rest of this paper, I will show that *mau* ‘want’ can also be used to mark the future, and I will investigate two ways in which it differs from the regular future marker *bakal*: (i) it behaves like a (weak) PPI, and (ii) it introduces a presupposition that its subject is disposed to causing the future event. In Section 2, I present the use of *mau* ‘want’ as a future marker. In Section 3, I show that its distribution empirically

² My consultant never uses the future marker *akan* except in formal contexts when she would speak in the standard rather than in the colloquial dialect. Therefore I will not discuss *akan* in the present paper. This is also an important difference with Copley (2010), who discusses *mau* as opposed to *akan*.

patterns with that of weak PPis. In Section 4, I discuss the requirement on the subject of future *mau*, namely its subject needs to be disposed at the time of evaluation to causing the future event. Finally, in Section 5, I propose an analysis to account for the similarities as well as the differences between desire and future *mau*. Section 6 concludes.

2. Future *mau*: ‘want’ as ‘will’

Colloquial Jakarta Indonesian has two verbs to express a ‘desire’ meaning, *pengen* and *mau*. When the meaning of ‘want’ is targeted, both can be used interchangeably, with no noticeable difference in their interpretations. This is illustrated in the following:

(5) Context: What job did you want to do when you were little?

Aku dulu pengen/mau jadi dokter.

I before want become doctor

‘I wanted to be a doctor.’

However, *mau* can also be used as a proximate future marker, without expressing the meaning of desire, and thus eliminating the need for an attitude holder. This use of *mau* is what I call the future *mau*.³

(6) Context: Two people are talking about whether the volcano will erupt or not.

Gunung=nya mau meletus..

Volcano.DEF FUT erupt

‘The volcano is about to erupt.’

³ I will be glossing future *mau* as FUT and desire *mau* (also spelled out as *pengen*) as ‘want’.

In the example above, the use of *mau* is not that of desire *mau*, since volcanos cannot have desires, and this is not a personification of the volcano. It thus seems that future *mau* no longer retains the desire component of ‘want’. Indeed, given that future *mau* does not express desires, there is no need for an attitude holder capable of having such desires. In other words, as we saw above, future *mau* can be used with inanimate subjects to which we cannot attribute intensionality or desires:

(7) Context: The food has been in the oven for a long time. My partner warns me:

Makanan=nya mau gosong!

Food-DEF FUT burn

‘The food is about to burn!’

The example above is perfectly felicitous in a situation where *mau* is expressing a proximate future, and it does not imply any metaphorical or “funny” readings attributing desires to its inanimate subject.

As a matter of fact, *pengen* perfectly contrasts with future *mau*, since its distribution is exactly what one would expect ‘want’ to be in a language, being infelicitous if we target a future meaning which cannot be re-analyzed as a desire. In the example below, where the subject is inanimate, and thus *mau* does not have a parse as a desire verb, *pengen* is infelicitous:

(8) Context: The book is at the edge of the bookshelf, I am afraid it will fall.

a. Buku=nya mau jatuh.

book-DEF FUT fall

‘The book is about to fall.’

b. # Buku=nya pengen jatuh.

book-DEF want fall

The use of *pengen* in (8b) has the same effect as its English counterpart would have, giving rise to an implausible metaphorical reading, where desires are being attributed to the book.

As expected by a future marker, *mau* also has future-in-the-past readings, such as in the following example where it is equivalent to ‘was about to’:

(9) Context: I was going to take the train yesterday to visit my friend in New York, but I missed it. I was at the train station on time but right when I was boarding I saw an old friend who called my name and we started chatting. I am explaining what happened to my friend in New York:

Ada yang mangil nama-ku waktu aku mau naik kereta.

EXIST REL.CL call name-POSS.1SG time I FUT board train

‘There’s someone who called my name when I was about to board the train.’

Is there one single under-specified lexical entry for *mau*, which sometimes gives rise to its desire meaning and sometimes to its future meaning? Achieving that is appealing, especially given that desire verbs are inherently future-oriented. I will partly account for the fact that it is a desire verb that turned into a proximate future by proposing a shared Logical Form for both desire and future *mau*. However, the two will need to have minimally different lexical entries. Where future *mau* differs from desire *mau* is (a) in not requiring an animate subject, and (b) in requiring that there are “signs” of the future event (more on this in Section 4). For these reasons, we do ultimately need two distinct lexical entries, although the two are of the same type semantically, and share the same Logical Form.

Positing two distinct lexical entries would also explain why both desire and future *mau* can be used in the same context, one being asserted and the other being

negated without generating a contradiction. As shown below, we can negate desire *mau* and asserting future *mau* in (10a) without any contradiction arising:

(10) Context: We are at a party, but it's getting late. I need to leave.

a. Sebenarnya aku **nggak mau**, tapi aku **mau** pulang sekarang ya.

Actually I NEG want but I FUT go-home now ok

'I don't actually want to, but I will go home now.'

b. Sebenarnya aku **nggak pengen**, tapi aku **mau** pulang sekarang ya.

Actually I NEG want but I FUT go-home now ok

'I don't actually want to, but I will go home now.'

c. # Sebenarnya aku **nggak pengen**, tapi aku **pengen** pulang sekarang ya.

Actually I NEG want but I want go-home now ok

d. # Sebenarnya aku **nggak mau**, tapi aku **pengen** pulang sekarang ya.

Actually I NEG want but I want go-home now ok

It's also possible to negate *pengen* and assert future *mau*, as in (10b), and again, no contradiction arises. As a control, we can see that negating and asserting *pengen* gives rise to a contradiction, since in (10c) the speaker has contradictory desires.

As a reviewer points out, however, the fact that we can assert one *mau* and negate the other is not necessarily knock-down evidence against a single lexical entry shared by both. We do, after all, know that the same modal with two different flavors can be asserted in one flavor and negated in another without contradiction. It is thus possible that here, too, we are asserting one *mau* quantifying over some set of worlds, and negating another 'flavor' of *mau* quantifying over a different set of worlds. Ultimately, this is the intuition I want to express as well, and for this reason, I will treat future and desire *mau* as similarly as possible. For example, I will propose that

they share the same position at the Logical Form, thus explaining why they cannot be stacked, i.e., why *mau mau* is ungrammatical:

(11) Context: Lia has two kids, Budi and Karina. They are very jealous of each other, so every time she buys something for Budi, Karina wants a gift too and vice versa. It's Budi's birthday, so she bought a gift for him, but forgot to buy something for Karina. Right before she gives the gift, Lia says to her husband:

- a. Karina bakal mau juga hadiah=nya.

Karina FUT want also present.DEF

'Karina will also want the present.'

- b. * Karina mau mau juga hadiah=nya.

Karina FUT want also present.DEF

However, I cannot propose a single lexical entry for both, since they do not only differ with respect to which worlds they quantify over. In other words, they cannot be different flavors of the same verb. As will become clear in the rest of the paper, future *mau* not only does not require an animate subject, but also further imposes a 'dispositional requirement' as I will call it on its subject. I will ultimately model this as a presupposition that only future *mau* has.

Diachronically, it is perfectly plausible that one has been derived from another, since this is a well attested cross-linguistic pattern (e.g., Bybee and Pagliuca (1987); Bybee and Dahl (1989); Nesselhauf (2012); Heine (2017) a.o.). What is interesting in Colloquial Jakarta Indonesian is that synchronically, the same verb, namely *mau*, can mean 'want' like *pengen* and 'will' like *bakal*.

Let us go back for a moment to (10). One may wonder: why is (10d) contradictory, where *mau* is being negated and *pengen* asserted? If *mau* can have both a desire and a future meaning, why do we not save the structure in (10d) by

interpreting NEG MAU as NEG FUT, while asserting *pengen* ‘want’? This would be a perfectly coherent meaning of the sort ‘I’m not actually going to, but I want to go home now’ (in a context where I stay to do someone a favor against my will for example).

The reason why this parse is unavailable is because future *mau* (but crucially not desire *mau* as we can see in (10a)) behaves like a PPI. In the next session, I will present the relevant data, showing that future *mau* empirically patterns with weak PPIS.

3. A puzzle with negation: future *mau* as a PPI

One of the novel empirical generalizations made in this paper is that future *mau* behaves like a PPI. It cannot be directly negated, and adding the colloquial negation *nggak* before *mau* can only give rise to a ‘not want’ meaning rather than a ‘will not’ one. This is illustrated with the following example, where the context is asking a question about the future, and yet the only possible interpretation of *nggak mau* is negating desire *mau*.

(12) Context: Will you be at the concert tomorrow?

Aku *nggak mau* ke konser=nya.

I NEG want to concert-DEF

‘I don’t ✓want to / ✗will not go to the concert.’

I should note that, since the dialect studied in this paper is Colloquial Jakarta Indonesian, I will only be considering the negation *nggak*. My consultant only uses the *tidak* negation in formal contexts.

Interestingly, when the negation *nggak* is added to *mau* the sentence regains the ambiguity between a present and a past tense interpretation that we see with bare verbs. In other words, a sentence with *nggak mau* behaves as if there is no future marker. This naturally follows if future *mau* is a PPI, which only leaves to the negation *nggak* the option of applying to desire *mau*. Since the latter is bare, it is ambiguous between a present and a past tense interpretation at the absence of a future marker. This is illustrated below in (13):

(13) Context: There is a faculty trip being organized, and no students are allowed to attend.

Nggak ada mahasiswa yang mau pergi.

NEG EXIST student REL.CL want go

‘There is no student who ✓wants / ✓wanted to / ✗will go.’

Given that the negation only targets desire *mau*, it is naturally incompatible with inanimate objects, which cannot have desires as in (14). This is expected if [NEG future *mau*] is an unavailable parse.

(14) Context: We are preparing the room for an exam, and the clock is not working.

*Jam=nya nggak mau nunjukin waktu yang benar.

clock-DEF NEG want show time REL.CL correct

Intended meaning: ‘The clock did not show the correct time.’

Is future *mau* only incompatible with a negation directly preceding it? No, on a par with other PPIS, future *mau* is incompatible with downward entailing quantifiers. We already saw this for the negative quantifier in (13), but we also illustrate it below for NOT ALL:

(15) Context: There is a school trip being organized, and all students are looking forward to attending. Unfortunately, there are not enough places for all of them.

Nggak semua murid ✓bakal / ✗mau pergi

NEG all student FUT want go

‘Not all students will go.’

How, then, can you negate the future? In all these environments, where we have a negation, a negative or a downward entailing quantifier, we would use the regular future marker *bakal*. The latter can be negated, and is perfectly felicitous in the environments above, where *mau* is infelicitous. Indeed, as we can see in (15) *bakal* is used with the quantifier NOT ALL.

If future *mau* is a PPI, we should ask what kind of a PPI it is. Let us follow the literature (e.g., Spector (2014)) in calling negation and other environments that make a PPI infelicitous, such as downward-entailingness, *antilicensers*. The strength of a PPI can be calculated based on whether or not it cares about the clause boundaries between its antilicenser and itself. Borrowing the terminology of Spector (2014), let us call a PPI *weak* if it tolerates an antilicenser across a clause boundary, and *global* if it does not tolerate any antilicenser.⁴ What kind of a PPI is future *mau*?

If the antilicenser is in a higher clause and future *mau* is separated with a clause boundary from it, then future *mau* is felicitous. Indeed, in the following example, if we negate the verb *yakin* ‘believe’ in the higher clause, we can still use *mau*, even with an inanimate subject. This suggests that *mau* is felicitous in this context even though it cannot be re-analyzed as desire *mau*.

⁴ I leave the question of whether the number of antilicensers matters, and more specifically whether the PPI is licensed with an even but not with an odd number of antilicensers, open for future research.

(16) Context: My colleague and I are preparing the room for an exam. I think the clock is about to stop, but my colleague reassures me it will not stop during the exam.

Aku nggak yakin kalo jam=nya mau mati.

I NEG believe COMP clock-DEF FUT stop

‘I don’t believe that the clock will stop.’

If we assume that *yakin* ‘believe’ is a neg-raising verb, then NEG *yakin...mau* should be equivalent to *yakin...NEG mau* at the level of the Logical Form. This would indicate that even though we cannot negate future *mau* directly, we can do so indirectly.

However, it is not clear that *yakin* ‘believe’ is indeed a neg-raising verb, and thus that we can indirectly negate future *mau*. *Yakin* is sometimes translated as ‘certain’, which is not neg-raising, since ‘not certain that x’ is not usually interpreted as ‘certain that not x’. What is more, in the following context *yakin* does not behave like a neg-raising verb, since (17a) conveys a very different meaning than (17b). Uttering (17a), we are uncertain about whether the horse will win or not, while uttering (17b) we are certain that the horse will lose.

(17) Context: I own a horse, and my horse is running a race today. During the time of the race, I have to be in an airplane, so I will not be able to see who’s leading. In the airplane, while the race is taking place, I say:

a. Aku nggak yakin kuda-ku bakal menang.

I NEG certain horse-POSS.1SG FUT win

‘I’m not sure that my horse will win.’

b. Aku yakin kuda-ku nggak bakal menang.

I certain horse- POSS.1SG FUT NEG FUT win

‘I’m sure that my horse won’t win.’

However, in other cases, when *yakin* is translated as ‘believe’ it behaves like a neg-raising verb, as in the example below.

(18) Context: Is Karina a good or a bad student?

a. Saya nggak yakin kalo dia rajin.

I NEG believe COMP she studious

‘I don’t believe that she is studious.’

b. Saya yakin kalo dia nggak rajin.

I believe COMP she NEG studious

‘I don’t believe that she is studious.’

Given that the data point to different directions, I cannot conclude that *yakin* is a neg-raising verb, although it is possible that whenever it is translated as ‘believe’ it behaves like a neg-raising verb.

That being said, future *mau* still behaves like a weak PPI, since the presence of a negation in a higher clause as in (16) does not antilicense it. The same is true for other kinds of antilicensers, like negative quantifiers. Future *mau* is compatible with a negative quantifier if the latter is on a higher clause. As in (16), I control for the desire reading of future *mau* by using an inanimate subject, which cannot have desires.

(19) Context: There is a volcano, but it hasn’t been active for years.

Nggak ada yang percaya kalo gunungnya mau melatus.

NEG EXIST REL.CL believe COMP volcano-DEF FUT erupt

‘There is no one who thinks that the volcano will erupt.’

As we can see above, when a negative quantifier is separated by a clause boundary from future *mau*, the latter is felicitous, even when the subject is inanimate. One could wonder whether there are in fact two clause boundaries between the negative quantifier and future *mau* in (19), given that there are both *yang* and *kalo*.

Investigating whether *yang* introduces a clause boundary is beyond the scope of this paper, however, at least for the purposes of future *mau* licensing, *yang* does not seem to introduce a clause boundary. Indeed, as we had seen in (13), repeated below as (20) for reference, a negative quantifier antilicenses future *mau*, and in that case too *yang* did not count as introducing a clause boundary.

(20) Context: There is a faculty trip being organized, and no students are allowed to attend.

Nggak ada mahasiswa yang mau pergi.

NEG EXIST student REL.CL want go

‘There is no student who ✓wants / ✓wanted to / ✗will go.’

I will state this as an open puzzle, calling it the ‘*yang* generalization’:

(21) The *yang* generalization: the relative clause marker *yang* does not count as introducing a clause boundary for the purposes of satisfying the PPI-hood of future *mau*. If future *mau* and a c-commanding negation are only separated by *yang*, future *mau* is not licensed.

We thus see that future *mau* behaves like a weak PPI, since it tolerates antilicensers separated by a clause boundary. This also refutes a simple theory which would suggest that there is some surface incompatibility of future *mau* with negation or negative quantifiers. Indeed, the distribution of future *mau* closely follows the distribution of weak PPIs like English *something*. The latter cannot be interpreted in the scope of a clausemate negation but can be interpreted within the scope of a negation across a clause boundary (see Szabolcsi (2004); Nicolae (2012) a.o.).

(22) John didn’t say something.

Can mean: there is something that John did not say.

Cannot mean: John said nothing.

(23) I don't think that John said something.

The distribution of future *mau* perfectly aligns with that of a weak PPI like *something*.

Future *mau*, like *something*, is compatible with ellipsis, suggesting once again that there is no surface incompatibility with the negation *nggak*.

(24) Context: I think the clock is about to stop, but my colleague is less certain.

Aku nggak tau jam=nya mau mati atau nggak.

I NEG know clock-DEF FUT stop or NEG

'I don't know whether the clock will stop or not.'

I should note, however, that it is unclear whether future *mau* is indeed being elided here, since the speaker cannot overtly pronounce it:

(25) Context: I think the clock is about to stop, but my colleague is less certain.

?? Aku nggak tau jam=nya mau mati atau nggak mau mati.

I NEG know clock-DEF FUT stop or NEG FUT stop

Intended: 'I don't know whether the clock will stop or will not stop.'

If anything, it seems that what is being elided is the regular colloquial future marker *bakal*, even though it is not perfectly pronounceable either. The speaker does, however, report a contrast between pronouncing *bakal* as in (26) and pronouncing *mau* as in (25). The latter is less felicitous.

(26) Context: I think the clock is about to stop, but my colleague is less certain.

? Aku nggak tau jam=nya mau mati atau nggak bakal mati.

I NEG know clock-DEF FUT stop or NEG FUT stop

'I don't know whether the clock will stop or will not stop.'

Eliding simply the VP is ungrammatical, since the sentence with two disjoined pronounced VPs is ungrammatical:

(27) Context: I think the clock is about to stop, but my colleague is less certain.

* Aku nggak tau jam=nya mau mati atau nggak mati.

I NEG know clock-DEF FUT stop or NEG stop

Intended: 'I don't know whether the clock will stop or not stop.'

To recapitulate, future *mau* is compatible with a negation when elided, as in (28) below, although it is unclear (and would depend on the identity condition we assume for ellipsis) whether *mau* or *bakal* is being elided in this case.

(28) Context: There is a volcano, and people are discussing whether it will explode. An expert is very uncertain.

Gunung=nya mungkin mau melatus, mungkin juga nggak.

Volcano-DEF possible FUT erupt possible also NEG

'The volcano might erupt, but it might also not.'

Once again, the ellipsis facts follow the distribution of PPis like *something*.

(29) May said something, but John didn't.

I argued that there is no surface incompatibility between the negation *nggak* and future *mau*, but I did not so far show any examples where '*nggak* future *mau*' directly surfaces. I argue that this is possible with metalinguistic negation, which does not affect truth conditions like regular negation. In the example below, *nggak mau* surfaces, with contrastive stress (marked by capitalization) on *NGGAK* and on *UDAH*. Crucially, the future meaning conveyed by *mau* is not directly negated despite the appearance of the surface string *nggak mau*.

(30) Context: A precious old library book is about to fall. I say "Bukunya mau jatuh!" (The book is about to fall!). My friend answers:

Nggak, itu buku=nya *NGGAK* mau jatuh, itu buku=nya *UDAH* jatuh.

No, this book-DEF NEG FUT fall this book-DEF already fall

'No, this book is not about to fall, it already fell.'

The same can be said for metalinguistic negation with weak PPIs like something (Nicolae 2012):

(31) John didn't say SOMETHING, he said EVERYTHING.

To summarize this section, we saw that future *mau* cannot be directly negated or be in the same clause with an antilicenser, such as a negative or a downward entailing quantifier. It patterns with weak PPIs, like *something*, in that it can tolerate an antilicenser across a clause boundary. It also behaves similarly to PPIs with respect to ellipsis, and its compatibility with metalinguistic negation. It would be natural to account for this on a par with other PPIs, which parallel to Negative Polarity Items have been argued to give rise to contradictory meanings when their meaning is exhausted in the scope of an antilicenser (see, e.g., Nicolae (2012); Iatridou and Zeijlstra (2013); Spector (2014); Nicolae (2017) a.o.). However, in this case, it is not clear what the alternatives would be. So, for the purposes of this paper, I characterize future *mau* as a PPI, leaving an analysis of its PPI-hood for future research. In the next Section, I will explore another interesting novel generalization about the meaning of future *mau*, namely that its subject needs to show signs that it will cause the future event.

4. The dispositional requirement

We have seen that *mau* can be used as a future marker, and that in this capacity it has the distribution of a weak PPI. A natural question to ask would then be: can we replace any positive instance of the regular future marker *bakal* with *mau*? In fact, we will argue that not all *bakals* can be *maus*, even in positive environments, where there is

no antilicenser. To the best of my knowledge, this property of future *mau* has not been described yet. I will propose a generalization, according to which there is a dispositional requirement on the subject of future *mau*, which I will argue is a presupposition. I will characterize this dispositional requirement borrowing the notion of dispositional causation from Copley (2018) and arguing that the subject of future *mau* needs to be disposed at the time of evaluation towards *mau*'s prejacent proposition.

The intuition of my consultant is that for future *mau* to be licensed there need to be “signs” for the future event. For instance, in the following example, the water boiling indicates that it will overflow.

(32) Context: I am cooking pasta. The water is boiling, and the lid is shaking.

Air=nya mau meluap.

Water-DEF FUT overflow

‘The water is about to overflow.’

To further illustrate this, consider the scenario in (8a), repeated below as (33) for reference:

(33) Context: The book is at the edge of the bookshelf, I am afraid it will fall.

Buku=mya mau / bakal jatuh.

book-DEF FUT fall

‘The book is about to/will fall.’

This is a scenario where future *mau* would be perfectly licensed, just like the regular future marker *bakal*. In this case, the book being at the edge of the bookshelf is what counts as the “sign” that licenses future *mau*. In other words, the book is in a state that may result in its falling. Intuitively, it seems that the subject is in a state (e.g., at the edge of the bookshelf) that can directly cause the future event (e.g., its falling).

If the subject is not in such a state, i.e., if there is no “sign” that the book is about to fall, then future *mau* is infelicitous, and *bakal* would be used instead.

(34) Context: The book is well-secured in the middle of the bookshelf. However, a magician in the room who wants to show off his skills says:

a. Buku=nya bakal / # mau jatuh.

book-DEF FUT fall

‘The book will fall.’

b. Tanpa diapa-apain buku itu bakal / # mau jatuh.

without nothing-done book DEM FUT fall

‘Without any reason, that book will fall.’

I propose to characterize this requirement governing the distribution of future *mau* as a dispositional requirement. More specifically, in the next section I will argue that the subject should be the holder of some state *e* at the time of evaluation, which can directly cause a state *e'*, instantiating the future event. Crucially, future *mau* is not licensed if there is no such state. For example, the book is not disposed to falling in (34), because it’s well secured in the middle of the bookshelf. Thus, future *mau* is not licensed. However, if the book is at the edge of the bookshelf as in (33), then future *mau* becomes perfectly felicitous. Interestingly, in the example below, clouds are the “signs” of rain licensing future *mau*, but this time with an expletive subject:

(35) Context: I look out the window, and there are dark clouds outside.

(Ini) ✓bakal / ✓mau hujan.

EXPL FUT rain

‘It’s about to rain.’

Supposing the proposed generalization that there is a dispositional requirement on the subject future *mau* is on the right track, I need to assume that in the example above

where it looks like there is an expletive subject there is in fact some silent subject, e.g., ‘the weather’, at the Logical Form. This is not implausible for rain sentences, and in fact, in English too, we can use a dummy subject with weather verbs even when we use an intensional verb, which would require not only a subject but also an attitude holder (Norvin Richards, p.c.):

(36) Context: There are some clouds.

- a. It’s thinking about raining.
- b. It’s trying to decide whether to rain or not.

Is this dispositional requirement a presupposition or is it part of the asserted content of future *mau*? It is hard to directly test this, since future *mau* is a PPI and cannot be directly negated. However, luckily, it is a weak PPI, and therefore we can test whether the presupposition projects over a negation in the higher clause. The following example suggests that it does.

(37) Context: Karina, Ali and Mia are classmates. Karina is very bad at math, but took intensive private classes over the summer and has now become great at math! Ali, her best friend, knows this. But Mia, who is not their friend still thinks that Karina is bad at math. On the first day of school she says that Karina will fail the class. Ali objects to defend her:

Aku nggak yakin kalo Karina √bakal / Xmau gagal!

I NEG believe COMP Karina FUT fail

‘I don’t believe that Karina will fail!’

The reason why future *mau* cannot felicitously be used in the example above cannot be the negation, since there is a clause boundary, and as already shown above, antilicensers in a higher clause do not affect the licensing of future *mau*. (37) is instead a presupposition failure, since the presupposition that Karina is in a state that can

directly cause her to fail at math is no longer true for Ali, who knows that she is now good at math. In other words, Ali has no “sign” that Karina will fail, the presupposition projects over the negation of the higher clause.

Another way to indirectly negate future *mau* is when it is elided under negation. In that case too, its presupposition projects. In the following example, what is targeted by the possibility modal *mungkin* ‘possible’ is the assertive content of future *mau*, not the dispositional requirement on its subject. In other words, the speaker does not convey that there is a possibility that the book is not in a state that can cause its falling; rather, what the speaker is asserting is that, despite being in that state, there is a chance that the book will not fall.

(38) Context: The book is at the edge of the bookshelf.

Buku=nya mau jatuh itu, tapi mungkin juga nggak.

book-DEF FUT fall DEM but possible also NEG

‘The book is about to fall, but it’s also possible it won’t.’

In this Section, I examined the ways in which the distribution of future *mau* is different than that of the regular future *bakal*. I proposed the generalization that there is a dispositional requirement on the subject of future *mau*, which needs to be in a state (at the time of evaluation) that can directly cause its prejacent proposition. I further argued that this requirement is a presupposition. In what follows, I will propose a lexical entry for future *mau* using the notion of dispositional causation (Copley 2018), while also connecting it to argument structure of desire *mau*. Future *mau* and desire *mau* will be shown to occupy the same position at the Logical Form, explaining the fact that they cannot be used together. They also share their future-orientation, which explains why it is a desire verb (and not some other verb) that synchronically behaves as a proximate future.

5. ‘Mau’ as the dispositional ‘will’

I propose that future *mau* shares its argument structure with desire *mau*, i.e. want, while also functioning as a future marker. It is thus similar to both *want* and *will*. Diachronically, the future in many languages is connected to desire verbs (see e.g., Bybee and Pagliuca (1987); Bybee and Dahl (1989); Nesselhauf (2012); Heine (2017) a.o.). This is no accident since desires are typically future-oriented, being satisfiable only in the future.⁵ Even past tense desires are future-oriented with respect to the past moment, similarly to the future-in-the-past readings of future *mau*. It is thus very tempting to propose a unified entry for future *mau* and desire *mau*. There are however a few differences between the two that make the task harder. Desire *mau* like *pengen* and *want* quantifies over deontic alternatives of an animate attitude holder. Future *mau*, on the other hand, under a *will* as modal approach (e.g., Reichenbach (1947), Prior (1967), Thomason (1970, 1984), Klein (1994), Huddleston (1995), Enç (1996), Condoravdi (2002), Copley (2002, 2008), Klecha (2014), Giannakidou & Mari (2018), and many more), quantifies over metaphysically accessible worlds, independently of any animate attitude holder.⁶ That being said, it seems plausible that

⁵ It is interesting to explore the difference between wishes (as in *I wish*) and desires (as in *I want*) in that sense, especially in counterfactuals. I leave that aside for the purposes of this paper, focusing on *want* vs. *will* rather than *wish*.

⁶ I leave the exact nature of this accessibility relation (i.e., whether they are indeed *metaphysically* accessible worlds or constrained in some other way) for Colloquial Jakarta Indonesian for future research.

the ‘buletic’ component of desire *mau* has been turned into the dispositional requirement of future *mau*. In other words, the quantification over buletic alternatives of an attitude holder gets bleached to quantification over accessible worlds. What used to be the attitude holder is now the subject that is disposed towards the prejacent of future *mau*. A subject being disposed towards a proposition is the closest we can get to ‘desire’ at the absence of an animate attitude holder. I will thus propose a common argument structure for both desire and future *mau*, justifying the use of *mau* as a future marker. I will still have to propose two lexical entries to account for the difference in accessibility relation, and in the dispositional requirement which I will formulate as a presupposition on future *mau*.

In line with Copley (2002, 2010), I analyze future *mau* as the dispositional ‘will’, suggesting that its subject is disposed to causing the eventuality of the verb. Copley (2002) discusses uses of dispositional ‘will’ in English, like the following example, which has a generic, dispositional reading:

(39) Dogs will eat doughnuts. Copley (2002): (188)

Dispositional reading: If you give a dog a doughnuts, the dog will eat it.

Copley (2010) also notes that (Standard) Indonesian future *mau* specifically can be used in dispositionals:

(40) Context: Does Budi eat fish?

Budi mau makan ikan.

Budi FUT eat fish

‘Budi will eat fish.’

Copley (2018): (4a)

My examples are both similar and dissimilar to these, since the subject of future *mau* is disposed to causing the future event (e.g., the book is disposed to falling by virtue of being at the edge of the bookshelf), but without a generic reading. I should also

highlight again that I am studying Colloquial Jakarta Indonesian, which does behave differently than Standard Indonesian with respect to the default future marker (*bakal* for Colloquial Jakarta, *akan* for Standard Indonesian).

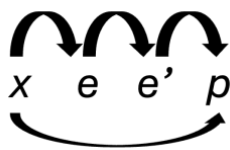
I am thus going to propose a different dispositional *will* for Colloquial Jakarta Indonesian, using the notion of dispositional causation, formalized in Copley (2018). What is required for future *mau* to be licensed is that its subject is disposed to causing its prejacent. Adapting Copley's (2018) notion of dispositional causation to future *mau*, I argue that its subject is disposed towards the prejacent proposition at the time of evaluation. More formally:

(41) The subject x of future *mau* is disposed towards the future proposition p at time t , represented by $\text{DISP}(x, p, t)$ if and only if:

- a. x is the holder of a state e at t
- b. e is a state that directly causes e' , ceteris paribus
- c. e' instantiates p
- d. x is disposed towards p at t

Modified from Copley (2018:32)

Figure 1. Schematic representation of (40).



Copley (2018:33)

I will thus say that the subject of future *mau* is disposed towards p at t if it is in a state that can directly cause another state, instantiating p . For example, in (8), repeated below as (42) for clarity, the subject of future *mau* is the book.

(42) Context: The book is at the edge of the bookshelf, I am afraid it will fall.

Buku=nya mau jatuh.

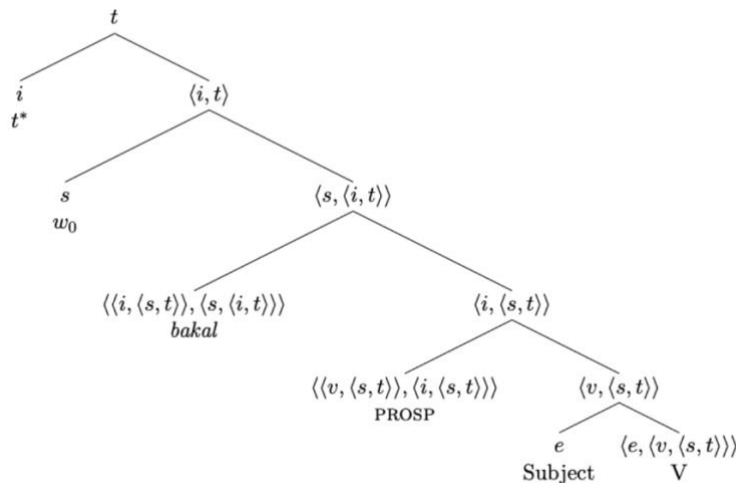
book-DEF FUT fall

‘The book is about to fall.’

The book is, at the time of utterance, in the state of being at the edge of the bookshelf, which can directly cause a state of falling, instantiating the future event. In other words, the state e the subject is in at the time of evaluation is what my consultant intuitively calls the “signs” for the future event.

Having defined what I mean by x is disposed towards p at t , I will now propose lexical entries for the regular future marker *bakal*, desire *mau*, and future *mau*. I will implement these in a three-valued logic system, giving truth, falsity, and undefinedness conditions when necessary. First, let me start with *bakal*, for which I adopt the *will* as modal approach (e.g., Reichenbach (1947), Prior (1967), Thomason (1970, 1984), Klein (1994), Huddleston (1995), Enç (1996), Condoravdi (2002), Copley (2002, 2008), Klecha (2014), Giannakidou & Mari (2018), and many more) as formalized in Bochnak (2019) but implemented in a three-valued system. The main idea behind the *will* as modal approach is that *will* is a modal rather than a tense, with its future meaning being contributed by a prospective aspect. I assume the following Logical Form for *bakal* sentences:

Figure 2. Logical Form schema of a *bakal* sentence.



Thus, *bakal* has the following truth conditions, like regular modal *will*.⁷ The future meaning is contributed by the prospective aspect, which requires the runtime of the event to be after the evaluation time.

(43) a. $\llbracket bakal \rrbracket = \lambda p_{i,st} . \lambda w . \lambda t .$

i. True if $\forall w' \in ACC(w, t)[p(t)(w') = 1]$

ii. False if $\forall w' \in ACC(w, t)[p(t)(w') = 0]$

b. $\llbracket PROSP \rrbracket = \lambda Q_{v,st} . \lambda t . \lambda w .$

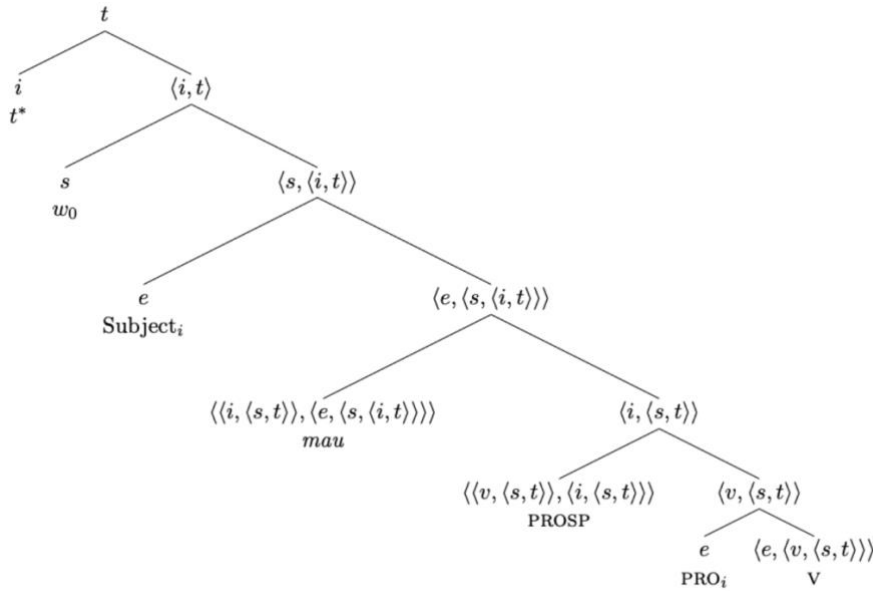
i. True if $\exists e[\tau(e) > t \& Q(e)(w)]$

ii. False if $\neg \exists e[\tau(e) > t \& Q(e)(w)]$

Adapted from Bochnak (2019): (13)-(15)

What about *mau*? Both future and desire *mau* will require a subject and an attitude holder respectively. They will therefore have a different Logical Form than *bakal*, as illustrated by the following:

Figure 3. Logical Form schema of a *mau* sentence.



⁷ There should also be an excluded middle presupposition to get the neg-raising behavior of *will*, but I abstract away from this for simplicity of exposition.

As shown in the Logical Form above, both future *mau* and desire *mau* have a future orientation thanks to the prospective aspectual marker, i.e., the same marker that also provides the future orientation of *bakal*. The difference with *bakal* is that *mau* also composes with a subject (animate in the case of desire *mau*, not necessarily animate in the case of future *mau*), which is co-referential with big PRO. The lexical entries for *mau* are as follows:

(44) a. $\llbracket \text{future mau} \rrbracket = \lambda p_{i,st} . \lambda x_e . \lambda w_s . \lambda t_i .$

i. True if $\forall w' \in ACC(w, t)[p(t)(w') = 1 \ \& \ DISP(x, p(t)(w'), t)]$

ii. False if $\forall w' \in ACC(w, t)[p(t)(w') = 0 \ \& \ DISP(x, p(t)(w'), t)]$

iii. Undefined otherwise

b. $\llbracket \text{desire mau} \rrbracket = \lambda p_{i,st} . \lambda x_e . \lambda w_s . \lambda t_i .$

i. True if $\forall w' \in BUL(x, w, t)[p(t)(w') = 1]$

ii. False if $\forall w' \in BUL(x, w, t)[p(t)(w') = 0]$

Future *mau* has a presupposition that its subject is disposed at the time of evaluation towards the prejacent proposition. We can see this because both truth and falsity conditions require that $DISP(x, p, t)$. If this condition is not met, we get undefinedness. Desire *mau*, on the other hand, uses its subject to calculate buletic alternatives, since these have to be pick out the set of worlds where the subject's desires at the time of evaluation t are met.⁸ To illustrate this, (42) would have the Logical Form in Figure 3, and the following truth conditions:

(45) $\llbracket (42) \rrbracket =$

*i. True if $\forall w' \in ACC(w_0, t *)$:*

⁸ Why does *pengen* not have a future *pengen* counterpart? This is an interesting question for future research, which could be related to the difference between *want* and *wish*, and other desire verbs more in general.

$\exists e [\tau(e) > t * \& fall(book, e, w')] \& DISP(book, \exists e [\tau(e) > t * \& fall(book, e, w')], t *)$

ii. False if $\forall w' \in ACC(w_0, t *)$:

$\neg \exists e [\tau(e) > t * \& fall(book, e, w')] \& DISP(book, \exists e [\tau(e) > t * \& fall(book, e, w')], t *)$

iii. Undefined otherwise

The sentence is true if the book is disposed (at the time of the utterance) to falling (e.g., by being at the edge of the bookshelf) and does fall in the future. The sentence is false if the book is disposed to falling and does not fall in the future, and undefined in all other cases.

I have modeled the dispositional requirement on future *mau* in terms of a presupposition, but I have not yet explained its PPI-hood. How can I account for the incompatibility of future *mau* (but not of desire *mau*) with clausemate antilicensers? Ideally, an explanation on a par with other PPis should be provided (Van der Wouden 2002; Szabolcsi 2004; Nicolae 2012; Iatridou and Zeijlstra 2013; Spector 2014; Homer 2015; Zeijlstra 2017), suggesting that at the presence of an antilicenser, if we exhaustify all stronger alternatives, we get a contradictory meaning. However, it is unclear what would count as a stronger alternative in the case of future *mau*. Intuitively, as a reviewer also points out, negating future *mau* gives rise to a contradiction between presupposed and asserted content (i.e., the subject is disposed to *p*, but *p* will not happen). Although intuitively appealing, this is not going to work, since there is no logical contradiction with having a disposition at the time of evaluation which does not lead to the future event. After all, all dispositions do not have to be realized in the future; a book at the edge of the bookshelf is very likely to fall, but doesn't have to (logically speaking) fall. The world would still be logically coherent if the book is disposed to falling and does not in the end fall. Thus, it is still unclear why future *mau* cannot be negated, and I leave this puzzle open for future research.

Although future *mau* cannot be negated, as expected, it is compatible with a negation in its scope as long as it is not c-commanded by another negation:

(46) Context: I had a dentist appointment, and he told me not to drink anything with color today except water.

Aku mau nggak minum teh nanti.

I FUT NEG drink tea later

‘I plan not to drink tea later.’

I thus proposed to account for the ambiguity of *mau* between a desire and a future *mau* through a common Logical Form, with two slightly different lexical entries. Desire *mau* behaves like a desire verb, quantifying over bulitic alternatives of an animate attitude holder. Future *mau*, on the other hand, does not require an animate attitude holder, but presupposes that its subject (be it animate or inanimate) is disposed at the time of evaluation towards its prejacent proposition. I formalized this using the notion of dispositional causation from Copley (2018).

6. Conclusion

This paper investigated the semantics of *mau*, which can sometimes be a desire verb sometimes a proximate future, in Colloquial Jakarta Indonesian. Two novel empirical generalizations were put forward. First, I showed that future *mau* behaves like a weak PPI in not tolerating clausemate c-commanding antilicensers. Secondly, I investigated a semantic requirement on the subject of future *mau*, which my consultant describes as there needing to be “signs” of the future event. This semantic requirement on its subject, as well as the fact that future *mau* is a proximate future, set it apart from

regular future marker *bakal*. I proposed to explain this difference, while also accounting for the similarity between future and desire *mau* by adopting a common Logical Form for both *maus*. Unlike desire *mau*, future *mau* only makes reference to its subject through a presupposition that it is disposed at the evaluation time towards the future proposition. Like desire *mau*, and unlike the regular future marker *bakal*, future *mau* composes with a subject. Like the regular future marker *bakal*, and unlike desire *mau*, future *mau* does not quantify over buletic alternatives, and thus does not require an animate attitude holder. I thus showed that future *mau* shares a Logical Form with desire verbs, while not requiring an animate attitude holder. The need for an attitude holder has been replaced by the dispositional requirement on its subject, which is disposed to, and metaphorically almost ‘wants’ to cause the future event. Many interesting puzzles remain open for future research, such as the explanation behind *mau*’s PPI-hood, and the diachronic connection between future and desire *mau*. Indonesian shows that desire verbs can *synchronically* have future usages, a change which is diachronically attested in many languages.

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Anastasia Tsilia

Department of Linguistics & Philosophy

Massachusetts Institute of Technology

32 Vassar St

Cambridge, MA, 02139

United States

tsilia@mit.edu